1)

```
SELECT COUNT(*) AS TotalPatients
FROM Patient;
```

100 % No issues found

Results Messages

	TotalPatients
1	25

The photo above shows the amount of rows that are populated in the Patient table.

2)

```
Select *
From Patient
Where PatientID = 5;

Delete From PatientAilment
Where PatientID = 5;

Delete From Patient
Where PatientID = 5;

Select *
From Patient
Where PatientID = 5;
```

00 % No issues found

Results Messages

PatientID	FName	LName	MName	DOB	Gender	Height	Weight

PatientID	FName	LName	MName	DOB	Gender	Height	Weight

(The beginning select statement shows nothing as I already deleted the data point when testing the script.)

The Photo above shows the select statements to see the data points and the delete statements for the data points then the verification that it is deleted.

3)

```sql
CREATE VIEW vw_PatientAilmentInfo AS
   SELECT
        p.PatientID,
        p.FName,
        p.LName,
        pa.AilmentID,
        pa.StartDate,
        pa.EndDate,
        pa.ProviderID
    FROM Patient AS p
    INNER JOIN PatientAilment AS pa
        ON p.PatientID = pa.PatientID;
go

SELECT * FROM vw_PatientAilmentInfo;
```

100 % No issues found

Results Messages

	PatientID	FName	LName	AilmentID	StartDate	EndDate	ProviderID
1	1	John	Smith	1	2018-01-10	NULL	1
2	1	John	Smith	4	2020-06-15	2020-09-01	3
3	2	Mary	Johnson	2	2019-03-22	NULL	2
4	2	Mary	Johnson	10	2021-04-01	2021-05-15	5
5	3	Robert	Brown	3	2017-08-12	NULL	4
6	3	Robert	Brown	9	2022-02-10	NULL	6
7	4	Patricia	Taylor	6	2020-11-05	NULL	7
8	4	Patricia	Taylor	15	2019-01-14	2019-06-30	8
9	6	Linda	Thomas	7	2018-09-03	NULL	11
10	6	Linda	Thomas	24	2020-10-10	2020-12-01	12
11	7	William	Jackson	8	2015-02-17	NULL	13
12	7	William	Jackson	23	2022-01-01	NULL	14
13	8	Barbara	White	10	2019-04-20	2019-06-01	15
14	8	Barbara	White	11	2023-01-05	2023-01-20	16
15	9	James	Harris	4	2018-07-11	NULL	17
16	9	James	Harris	18	2021-09-09	NULL	18

Shows the script that was made to join the two tables and the outcome of the two tables.

4)

```sql
SELECT
    v.PatientID,
    v.FName,
    v.LName,
    v.StartDate,
    v.EndDate,
    a.Ailment,
    a.AilDesc
FROM vw_PatientAilmentInfo AS v
INNER JOIN Ailment AS a
    ON v.AilmentID = a.AilmentID
WHERE v.AilmentID = 2;
```

100 % 8 0

Results Messages

	PatientID	FName	LName	StartDate	EndDate	Ailment	AilDesc
1	2	Mary	Johnson	2019-03-22	NULL	Diabetes Mellitus	Metabolic disorder characterized by high blood g...
2	20	Sandra	Hall	2016-06-06	NULL	Diabetes Mellitus	Metabolic disorder characterized by high blood g...

This is a 3 table join with 1 view and 1 table with a key value.

5)

a)

```sql
SELECT
    AVG(Weight) AS AveragePatientWeight
FROM Patient;
```

100 % 1 0

Results | Messages

	AveragePatientWeight
1	166.220833

i) This shows the average weight of all patients.

b)

```sql
SELECT
    MAX(Weight) AS HeaviestPatient,
    MIN(Weight) AS LightestPatient
FROM Patient;
```

100 % No issues found

Results | Messages

	HeaviestPatient	LightestPatient
1	210.8	135.6

i) This shows the Heaviest patient and the lightest patient in the same script

c)

```sql
SELECT
    ProviderID,
    UPPER(FName) AS FirstName,
    UPPER(LName) AS LastName
FROM Provider;
```

100 % No issues found

Results | Messages

	ProviderID	FirstName	LastName
1	1	JAMES	WILSON
2	2	EMILY	CARTER

i) This is a script that changes the names from looking like Bob to be all uppercase letters

d)

```sql
SELECT
    PatientID,
    AilmentID,
    StartDate,
    EndDate,
    DATEDIFF(DAY, StartDate, EndDate)
        AS DaysInTreatment
FROM PatientAilment
WHERE EndDate IS NOT NULL;
```

100 % No issues found

Results | Messages

	PatientID	AilmentID	StartDate	EndDate	DaysInTreatment
1	1	4	2020-06-15	2020-09-01	78
2	2	10	2021-04-01	2021-05-15	44
3	4	15	2019-01-14	2019-06-30	167

i) This shows how many days each patient has been in treatment.

e)

```sql
SELECT
    PatientID,
    CONCAT(FName, ' ',LName) AS FullName,
    DOB,
    Gender
FROM Patient;
```

100 % No issues found

Results | Messages

	PatientID	FullName	DOB	Gender
1	1	John Smith	1985-03-12	M
2	2	Mary Johnson	1990-07-25	F
3	3	Robert Brown	1978-11-02	M

i) This shows the Date of Birth as well as the Gender of each patient

6)

```sql
CREATE PROCEDURE usp_InsertProvider
    @FName          NVARCHAR(50),
    @LName          NVARCHAR(50),
    @ProviderTypeID INT
AS
BEGIN
    INSERT INTO Provider (FName, LName, ProviderTypeID)
    VALUES (@FName, @LName, @ProviderTypeID);

    SELECT *
    FROM Provider
    WHERE ProviderID = SCOPE_IDENTITY();
END;

EXEC usp_InsertProvider
    @FName          = 'Sarah',
    @LName          = 'Johnson',
    @ProviderTypeID = 1
```

100 % No issues found

Messages

```
Commands completed successfully.

Completion time: 2026-03-04T17:41:45.5984087-05:00
```

This script creates a procedure that inserts a provider directly into the table without having to go into the table itself.

7)

```sql
CREATE TABLE Appointment (
    AppointmentID   INT             not null,
    PatientID       INT             not null,
    ProviderID      INT             not null,
    ApptDate        INT             not null,
    ApptTime        INT             not null,
    ApptNotes       NVARCHAR(MAX)   null,

    CONSTRAINT PK_Appointment PRIMARY KEY (AppointmentID)
);

CREATE NONCLUSTERED INDEX IX_Appointment_PatientID
    ON Appointment (PatientID);
```

This script creates a table named Appointment with these rows and using int as input data and the row has NOT NULL then it is a required row.

8)

```sql
SELECT
    p.PatientID,
    p.FName,
    p.LName,
    COUNT(pa.AilmentID) AS TotalAilments
FROM Patient AS p
INNER JOIN PatientAilment AS pa
    ON p.PatientID = pa.PatientID
GROUP BY p.PatientID, p.FName, p.LName
ORDER BY TotalAilments DESC;
```

100 % No issues found

Results Messages

	PatientID	FName	LName	TotalAilments
1	1	John	Smith	2
2	2	Mary	Johnson	2
3	3	Robert	Brown	2
4	4	Patricia	Taylor	2

This script allows the user to see the patients that have the same number of Ailments together in one place.

9)

```sql
SELECT DISTINCT
    p.PatientID,
    p.FName,
    p.LName,
    p.DOB,
    p.Gender,
    p.Height,
    p.Weight
FROM Patient AS p
ORDER BY
    p.DOB       DESC,
    p.LName     ASC,
    p.FName     ASC;
```

100 % No issues found

Results Messages

	PatientID	FName	LName	DOB	Gender	Height	Weight
1	25	Mark	Lopez	1999-12-19	M	71.9	185.4
2	11	David	Thompson	1998-02-27	M	73.2	190.6
3	20	Sandra	Hall	1996-04-26	F	65.4	149.2
4	23	Paul	King	1994-02-10	M	68.5	165.7

This script shows the patients from youngest to oldest in descending order

10)

```sql
SELECT
    p.PatientID,
    p.FName         AS PatientFirst,
    p.LName         AS PatientLast,
    pr.ProviderID,
    pr.FName        AS ProviderFirst,
    pr.LName        AS ProviderLast
FROM Patient AS p
INNER JOIN PatientAilment AS pa
    ON p.PatientID = pa.PatientID
INNER JOIN Provider AS pr
    ON pa.ProviderID = pr.ProviderID;
```

100 % No issues found

Results Messages

	PatientID	PatientFirst	PatientLast	ProviderID	ProviderFirst	ProviderLast
1	1	John	Smith	1	James	Wilson
2	1	John	Smith	3	Michael	Reed
3	2	Mary	Johnson	2	Emily	Carter
4	2	Mary	Johnson	5	David	Patel
5	3	Robert	Brown	4	Sarah	Nguyen
6	3	Robert	Brown	6	Laura	Hernandez

This demonstrates the INNER JOIN function. It shows the patient and provider that shows both of the tables.

11)

```sql
SELECT
    p.PatientID,
    p.FName         AS PatientFirst,
    p.LName         AS PatientLast,
    pa.AilmentID,
    pr.FName        AS ProviderFirst,
    pr.LName        AS ProviderLast
FROM Patient AS p
LEFT OUTER JOIN PatientAilment AS pa
    ON p.PatientID = pa.PatientID
LEFT OUTER JOIN Provider AS pr
    ON pa.ProviderID = pr.ProviderID;
```

100 % No issues found

Results Messages

	PatientID	PatientFirst	PatientLast	AilmentID	ProviderFirst	ProviderLast
1	1	John	Smith	1	James	Wilson
2	1	John	Smith	4	Michael	Reed
3	2	Mary	Johnson	2	Emily	Carter
4	2	Mary	Johnson	10	David	Patel
5	3	Robert	Brown	3	Sarah	Nguyen
6	3	Robert	Brown	9	Laura	Hernandez

This shows the left outer join of the PatientAilment and the Provider Tables.

12)

```sql
SELECT
    pr.ProviderID,
    pr.FName        AS ProviderFirst,
    pr.LName        AS ProviderLast,
    pt.ProviderType,
    pa.PatientID,
    pa.AilmentID
FROM PatientAilment AS pa
RIGHT OUTER JOIN Provider AS pr
    ON pa.ProviderID = pr.ProviderID
RIGHT OUTER JOIN ProviderType AS pt
    ON pr.ProviderTypeID = pt.ProviderTypeID;
```

100 % No issues found

Results Messages

	ProviderID	ProviderFirst	ProviderLast	ProviderType	PatientID	AilmentID
1	1	James	Wilson	Physician	1	1
2	1	James	Wilson	Physician	13	20
3	3	Michael	Reed	Physician	1	4
4	3	Michael	Reed	Physician	14	24
5	5	David	Patel	Physician	2	10
6	5	David	Patel	Physician	16	6

This shows the right outer join of the Provider table and ProviderType table.